

CS208 Cloud Computing and Cloud Networking

Assignment 4: Managing your own cloud

Live Migration (Putting theory into practice)

Live migration is not simple in practice; furthermore, it is impossible to diagnose what caused the migration to abort without understanding the relevant configuration and compatibilities. Please complete the assigned readings below, answer the questions briefly, and then run your experiments.

Reading 1:

Configure Live Migration Link:

<https://docs.openstack.org/nova/pike/admin/configuring-migrations.html#section-configuring-compute-migrations>

1. What is an ephemeral disk? Note: ephemeral disk is an Openstack storage option.
2. What are the differences between a) Shared storage based live migration b) Block live migration c) Volume backed live migration
3. What causes block live migration to take more time?
4. Considering KVM-libvirt general and advanced configurations:
 - a. What do the parameters below dictate:
 - i. live_migration_downtime
 - ii. live_migration_downtime_steps
 - iii. live_migration_downtime_delay
 - b. Is post-copy the default setting?

Reading 2:

Live Migrate Instances Link:

<https://docs.openstack.org/nova/pike/admin/live-migration-usage.html>

5. This reading mentions: “block migration compatible Xen hypervisors must support _____ feature”. Fill in the blank, what is it?
6. What log file (path) should you inspect when a migration fails?
7. What is auto-convergence?

General Instructions:

- Follow the steps shown in the link [\[1\]](#) from “14.4 Building YourOwn OpenStack Cloud” onwards on Cloudlab and document your entire progress with screenshots and relevant information. Your documentation shall be submitted to iLearn. In step 3 of
- [1], In “**Number of compute nodes (at Site 1)**”, please use **two compute nodes**. Please use the **Mitaka** Openstack release in step 3.
-

[1] Cloudlab Openstack installation: <https://docs.cloudlab.us/openstack-tutorial.html>

Migration Setup:

1. Note down hostname for all the compute node, forexampleatcomputenode 1:

```
hostname  
Output: cp-1.vjain014-qv91206.kkprojects-pg0.apt.emulab.net
```

2. Check if passwordless access between **all compute nodes** is enabled from the root user. For example for cp-1 (repeat for cp-2)

```
sudo -s  
ssh cp-2.vjain014-qv91206.kkprojects-pg0.apt.emulab.net  
Output: cp-1.vjain014-qv91206.kkprojects-pg0.apt.emulab.net
```

3. Choose one of the compute node as an NFS server and install following package: (Only on one of the compute node)

```
apt-get install nfs-kernel-server  
service nfs-kernel-server restart
```

4. Update the “/etc/hosts” entry of **all compute nodes** to have an entry for “nfs-server” with IP address mapping. (I chose cp-1 as my NFS server)

```

128.110.96.81 cp-1.vjain014-QV91206.kkprojects-PG0.appt.emulab.net
192.168.0.1 ctl
192.168.0.5 cp-2
192.168.0.7 cp-1 nfs-server
127.0.0.1 localhost localhost localhost.vjain014-qv91206.kkprojects-pg0.appt.emulab.net
10.11.10.3 cp-2-flat-lan-1 cp-2-0
10.11.10.1 ctl-flat-lan-1 ctl-0
10.11.10.2 cp-1-flat-lan-1 cp-1-0
128.110.96.81 cp-1.vjain014-QV91206.kkprojects-PG0.appt.emulab.net apt081.appt.emulab.net

```

5. Run the following command at the NFS server (node chosen at step 3). Set the execute/search bit on the instances directory, this allows qemu to access the instances directory tree:

```
chmod o+x /var/lib/nova/instances
```

6. Run the following command at the NFS server (node chosen at step 3). Export /var/lib/nova/instances to the compute hosts. For example, add the following line to “/etc/exports”:

```
echo"/var/lib/nova/instances *(rw,sync,fsid=0,no_root_squash)">> /etc/exports

service nfs-kernel-server restart
```

7. After setting up the NFS server, mount the remote filesystem on **all compute hosts** (but server node first). Assuming the NFS server's hostname is “nfs-server” (at step 4), add this line to /etc/fstab to mount the NFS root:

```
echo"nfs-server:/ /var/lib/nova/instances nfs4 defaults0 0">> /etc/fstab
```

8. Complete NFS on **all compute hosts** by mounting the instances directory:

```
mount -a -v
```

9. Following things need to be done in all compute node:

In /etc/libvirt/libvirtd.conf:

```
listen_tls = 0
listen_tcp = 1
```

```
auth_tcp = "none"
```

In `/etc/default/libvirt-bin`:

Please note, -l is Lower case L.

```
libvirtd_opts="-l"
```

Restart libvirt service:

```
service libvirt-bin restart
```

10. Source all the required environment variables:

```
source /root/setup/admin-openrc.sh
```

11. Obtain the ID of the instance you want to migrate:

```
$ openstack server list
```

ID	Name	Status	Networks	Image Name
d1df1b5a-70c4-4fed-98b7-423362f2c47c	vm1	ACTIVE	private=a.b.c.d	...
d693db9e-a7cf-45ef-a7c9-b3ecb5f22645	vm2	ACTIVE	private=e.f.g.h	...

12. Determine on which host the instance is currently running. In this example, vm1 is running on HostB:

```
$ openstack server show d1df1b5a-70c4-4fed-98b7-423362f2c47c
```

Field	Value
...	...
OS-EXT-SRV-ATTR:host	HostB
...	...
addresses	a.b.c.d
flavor	m1.tiny
id	d1df1b5a-70c4-4fed-98b7-423362f2c47c
name	vm1
status	ACTIVE
...	...

13. Select the compute node the instance will be migrated to. In this example, we will migrate the instance to **HostC**, because **nova-compute** is running on it:

```
$ openstack compute service list
```

ID	Binary	Host	Zone	Status	State	Updated At
3	nova-conductor	HostA	internal	enabled	up	2017-02-18T09:42:29.000000
4	nova-scheduler	HostA	internal	enabled	up	2017-02-18T09:42:26.000000
5	nova-consoleauth	HostA	internal	enabled	up	2017-02-18T09:42:29.000000
6	nova-compute	HostB	nova	enabled	up	2017-02-18T09:42:29.000000
7	nova-compute	HostC	nova	enabled	up	2017-02-18T09:42:29.000000

14. Migrate the instance:

```
openstack server migrate d1df1b5a-70c4-4fed-98b7-423362f2c47c --live HostC --block-migration
```

Experiment

1. Login to Openstack Controller to access OpenStack CLI and check what hypervisor you have?
(Your output should look something like this)

```
root@ctl:~# source /root/setup/admin-openrc.sh
root@ctl:~# openstack server list
```

ID	Name	Status	Networks	Image	Flavor
a09da69a-0599-40ea-9001-8208b905faf4	mklm-1	ACTIVE	ext-net=128.110.155.134	xenial-server	manila-service-flavor
eb926a6f-5590-46d8-b791-c569faf82130	mklm-2	ACTIVE	ext-net=128.110.155.135	xenial-server	manila-service-flavor

```
root@ctl:~#
```

```
root@ctl:~# openstack hypervisor list
```

ID	Hypervisor Hostname	Hypervisor Type	Host IP	State
1	cp-1.assignmenttest.kkprojects-pg0.utah.cloudlab.us	QEMU	192.168.0.5	up

2. Using screenshots and by highlighting relevant details, such as instance ID and hostnames (from and to), depict your ability to carry out **live migration** from one compute node to another compute node on the same site. **Please do not perform Non-live migration.** You may run the command 'date' before and after each command.

3. Can you plot the migration time against the flavor of the instance migrated?

Grading

What is tested?

- You have to create a single Word or PDF document for both of the parts (reading + experiments).

Grade Distribution

- Reading 1 - 20 points
- Reading 2 - 20 points
- Experiment- 60 points